

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1165

Delaware City Plant 138 kV
September 18, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1165. The Interconnection Customer proposes a 139.2 MW solar photovoltaic facility interconnecting at the Delaware City Plant 138 kV substation in the Delaware area. The request was received on March 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,231,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MILL CTEEK to SOUTHPPOINT 138kV line	TOMBSTONE to BIXBY HILL 500kV	147.6	172	5.6
SUNSET LAKE to BROADVIEW 138kV	DENHAM SPRINGS to OTAY TAP 138kV line	122.4	195	19.8
DELLCREEK to KEENEY 138kV Line	RED LION ENERGY CTR to DELAWARE CITY PLANT SUBSTATION 138kV	110.6	288	29.2
SOUTHPPOINT to ELIZABETH LAKE 138kV	BIXBY HILL to TOMBSTONE 500kV Line	107.7	206	5.6
RED LION ENERGY CENTER to MAURICE 138kV	KEENEY to DELL CRK 138kV	105.4	226	50.5
DELL CRK to CHURCHS FERRY 138kV	RED LION ENERGY CTR to DELAWARE CITY PLANT 138kV	105.0	142	54.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MILL CTEEK to SOUTHPOINT 138kV	Rebuild line with higher-capacity conductor	\$3,715,000	100% (\$3,715,000)
SUNSET LAKE to BROADVIEW 138kV	Rebuild line with higher-capacity conductor	\$3,309,000	75% (\$2,478,000)
DELL CRK to KEENEY 138kV Line	Rebuild line with higher-capacity conductor	\$2,404,000	77% (\$1,856,000)
SOUTHPOINT to ELIZABETH LAKE SUBSTATION 138kV L	Reconductor line	\$573,000	100% (\$573,000)
RED LION ENERGY CTR to MAUICE 138kV	Reconductor line	\$1,043,000	90% (\$940,000)
DELL CRK to CHURCHS FERRY 138kV	Reconductor line	\$820,000	82% (\$669,000)
	Total allocated to project		\$10,231,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.